

(TR-104) PYTHON WITH AI/ML

Training Day 84 Report

Date: 7 May 2026

Topic Covered: Schema Synchronization, Data Retrieval Optimization & Backend Refactoring

The eighty-fourth day of training focused on optimizing data retrieval, synchronizing backend services with the latest MongoDB schema, enhancing search capabilities, improving compliance validation logic, and standardizing backend components for better maintainability.

- Optimized data retrieval mechanisms to ensure complete dataset visibility across backend services.
- Resolved record visibility limitations and improved handling of large datasets.
- Enhanced trailer, customer, payment, and detail retrieval tools for better scalability and performance.
- Updated backend formatter logic to align with the latest MongoDB schema structure.
- Added support for newly introduced customer, trailer, and payment-related fields.
- Ensured API responses accurately reflect database schema fields for frontend consistency.
- Refactored MongoDB aggregation pipelines to include additional fields required for frontend rendering and AI responses.
- Verified proper retrieval and mapping of customer and trailer information through aggregation workflows.
- Enhanced regex-based search functionality and improved query flexibility across customer lookup tools.
- Added support for both new and legacy schema fields to maintain backward compatibility.
- Expanded searchable attributes to improve search accuracy and user experience.
- Improved compliance validation logic using work permit and truck cab card information.

- Updated sorting and grouping mechanisms for structured customer and payment management.
- Standardized formatter outputs and optimized backend service structures for improved maintainability.
- Verified schema synchronization and consistency across all updated backend modules.
- Revised concepts related to MongoDB schema synchronization, aggregation pipelines, scalable data retrieval, backend optimization, backward compatibility, and compliance validation.

Overall, the training enhanced understanding of schema migration, backend optimization, aggregation pipeline design, scalable data management, and enterprise-level API development.